Vectors: Components and Operations

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Vector arithmetic**

Find the component form and magnitude of the vector from $P(1, 3)$ to $Q(5, -2)$.

$\vec{PQ} = \langle 4, -5 \rangle$, $|\vec{PQ}| = \sqrt{16 + 25} = \sqrt{41}$.

2 **[Foundational]** • **Target: Vector arithmetic**

Write the vector of magnitude 12 making an angle of $150°$ with the positive x-axis in component form (exact values).

$\langle 12\cos 150°, 12\sin 150° \rangle = \langle -6\sqrt{3}, 6 \rangle$.

3 **[Foundational]** • **Target: Applications**

A plane flies due east at 200 mph while the wind blows due north at 50 mph.

a Find the magnitude of the plane's resultant velocity, to one decimal place.

$\sqrt{200^2 + 50^2} = \sqrt{42500} \approx 206.2$ mph.

b Find the direction of the resultant, as an angle north of east to two decimal places.

$\theta = \tan^{-1}\left(\frac{50}{200}\right) \approx 14.04°$ north of east.

(a) $\sqrt{200^2 + 50^2} = \sqrt{42500} \approx 206.2$ mph.; (b) $\theta = \tan^{-1}\left(\frac{50}{200}\right) \approx 14.04°$ north of east.

4 **[Foundational]** • **Target: Applications**

Two forces $\mathbf{F}_1 = \langle 30, 0 \rangle$ N and $\mathbf{F}_2 = \langle 0, 40 \rangle$ N act on an object. Find the magnitude of the resultant force.

$|\mathbf{F}_1 + \mathbf{F}_2| = |\langle 30, 40 \rangle| = 50$ N.

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5 [Foundational] • Target: Applications

A boat aims to cross a river by heading due north at 4 m/s, but the current pushes it east at 1.5 m/s.

a Find the boat's resultant speed, to two decimal places. $\sqrt{4^2 + 1.5^2} = \sqrt{18.25} \approx 4.27$ m/s.

b If the river is 200 m wide, find how long the crossing takes. Only the northward component moves it across: $t = \frac{200}{4} = 50$ s.

c Find how far downstream (east) the boat lands. Drift $= 1.5 \times 50 = 75$ m.

(a) $\sqrt{4^2 + 1.5^2} = \sqrt{18.25} \approx 4.27$ m/s.; (b) Only the northward component moves it across: $t = \frac{200}{4} = 50$ s.; (c) Drift $= 1.5 \times 50 = 75$ m.

6 [Foundational] • Target: Position vectors and geometry

Given $A(-2, 5)$ and $B(4, -3)$, find the midpoint of $\overline{AB}$ and the vector $\vec{AB}$.

Midpoint: $\left(\frac{-2+4}{2}, \frac{5-3}{2}\right) = (1, 1)$. $\vec{AB} = \langle 6, -8 \rangle$.

7 [Foundational] • Target: Position vectors and geometry

Let $\mathbf{a} = \langle 2, 3 \rangle$ and $\mathbf{b} = \langle -4, 1 \rangle$. Find a vector $\mathbf{c}$ such that $\mathbf{a} + \mathbf{b} + \mathbf{c} = \langle 0, 0 \rangle$, and verify your answer.

$\mathbf{c} = -(\mathbf{a} + \mathbf{b}) = -\langle -2, 4 \rangle = \langle 2, -4 \rangle$. Check: $\langle 2, 3 \rangle + \langle -4, 1 \rangle + \langle 2, -4 \rangle = \langle 0, 0 \rangle$. ✓

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Position vectors and geometry

Express $\mathbf{v} = \langle 5, -12 \rangle$ in terms of the standard unit vectors $\mathbf{i}$ and $\mathbf{j}$, and find $|\mathbf{v}|$.

$\mathbf{v} = 5\mathbf{i} - 12\mathbf{j}$. $|\mathbf{v}| = \sqrt{25 + 144} = 13$.

9 [Intermediate] • Target: More applications

An airplane has an airspeed of 305 mph and flies on a heading $7.5°$ west of due north, while a 40 mph wind blows due east. Find the plane's resultant ground velocity vector, and its ground speed, to one decimal place (take east as the positive x-direction and north as the positive y-direction).

Plane's own velocity: $\langle -305\sin 7.5°,\ 305\cos 7.5° \rangle \approx \langle -39.8,\ 302.4 \rangle$. Adding the wind $\langle 40, 0 \rangle$: resultant $\approx \langle 0.2,\ 302.4 \rangle$. Ground speed $\approx \sqrt{0.2^2 + 302.4^2} \approx 302.4$ mph — very nearly due north, as intended.

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10 [Intermediate] • Target: More applications

Given $\mathbf{u} = \langle 4, -1 \rangle$, find a vector orthogonal to $\mathbf{u}$ with the same magnitude as $\mathbf{u}$.

Swap components and negate one: $\langle 1, 4 \rangle$. Check: $\mathbf{u} \cdot \langle 1, 4 \rangle = 4 - 4 = 0$ (orthogonal), and $|\langle 1, 4 \rangle| = \sqrt{17} = |\mathbf{u}|$.

11 [Intermediate] • Target: Finding direction angles

Find the direction angle of $\mathbf{v} = \langle -3, 3\sqrt{3} \rangle$, to the nearest degree.

$\theta = \tan^{-1}\left(\frac{3\sqrt{3}}{-3}\right)$ with reference angle $60°$, but since the vector has negative x and positive y (Quadrant II): $\theta = 180° - 60° = 120°$.

12 [Intermediate] • Target: Finding direction angles

A vector has magnitude 10 and direction angle $200°$. Write it in component form, rounded to two decimal places.

$\langle 10\cos 200°, 10\sin 200° \rangle \approx \langle -9.40, -3.42 \rangle$.

13 [Intermediate] • Target: Vector equations and unknowns

Find scalars c_1 and c_2 such that $c_1\langle 1, 0 \rangle + c_2\langle 0, 1 \rangle = \langle -3, 7 \rangle$.

Since $\langle 1, 0 \rangle$ and $\langle 0, 1 \rangle$ are the standard basis vectors, $c_1 = -3$ and $c_2 = 7$ directly.

14 [Intermediate] • Target: Parallel and equal vectors

Determine whether $\mathbf{u} = \langle 4, -6 \rangle$ and $\mathbf{v} = \langle -6, 9 \rangle$ are parallel, and if so, state whether they point in the same or opposite directions.

$\mathbf{v} = -\frac{3}{2}\mathbf{u}$ (check: $-\frac{3}{2}(4) = -6$ ✓, $-\frac{3}{2}(-6) = 9$ ✓). Parallel, with a negative scalar multiple, so they point in OPPOSITE directions.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Parallel and equal vectors • Cross-Lesson: 1.01 Functions and Function Notation

Find the value of k so that $\langle k, 6 \rangle$ is parallel to $\langle 4, 8 \rangle$.

Need $\frac{k}{4} = \frac{6}{8} = \frac{3}{4}$: $k = 3$.

16 [Challenging] • Target: Vectors in three real-world scenarios • Cross-Lesson: 1.02 Domain and Range

A hiker walks 3 km east, then 4 km north, then 2 km west. Find the magnitude of the hiker's total displacement from the starting point.

Net displacement: $\langle 3 - 2, 4 \rangle = \langle 1, 4 \rangle$. Magnitude $= \sqrt{1 + 16} = \sqrt{17} \approx 4.12$ km.

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17 [Challenging] • Target: Vectors in three real-world scenarios • Cross-Lesson: 1.03 Average Rate of Change

Three ropes pull on a ring with forces $\langle 10, 0\rangle$, $\langle -5, 8.66\rangle$, and $\langle -5, -8.66\rangle$ N. Find the net force, and explain what this implies physically.

Sum: $\langle 10-5-5,\ 0+8.66-8.66\rangle = \langle 0, 0\rangle$. The net force is zero, meaning the ring is in equilibrium (the three forces are $120°$ apart and balance exactly).

18 [Challenging] • Target: Vectors in three real-world scenarios • Cross-Lesson: 2.01 Right-Triangle Trigonometry

A crane hoists a load using two cables. One cable exerts a force of $\langle -200, 350\rangle$ N and the other exerts $\langle 200, 400\rangle$ N. Find the combined lifting force (the vertical component of the resultant), and explain why the horizontal components matter for the crane's structural design even though they cancel.

Resultant: $\langle -200+200,\ 350+400\rangle = \langle 0, 750\rangle$. The vertical (lifting) force is 750 N. Although the horizontal components cancel in the resultant, each individual cable still experiences its own horizontal tension, which the crane's frame must be strong enough to withstand even though the net horizontal effect on the load is zero.

19 [Challenging] • Target: Vectors in three real-world scenarios • Cross-Lesson: 2.02 Radian Measure and Arc Length

A swimmer aims to swim directly across a river at 1.5 m/s relative to the water, while the current flows at 0.8 m/s downstream. Find the swimmer's resultant speed relative to the shore, and the angle (from the intended straight-across direction) at which they actually drift, to the nearest degree.

Resultant speed $= \sqrt{1.5^2 + 0.8^2} = \sqrt{2.25 + 0.64} = \sqrt{2.89} = 1.7$ m/s. Drift angle $= \tan^{-1}\left(\frac{0.8}{1.5}\right) \approx 28°$ from the intended direction.

20 [Challenging] • Target: More applications • Cross-Lesson: 2.03 Graphs of Sine and Cosine Functions

Given vectors $\mathbf{u} = \langle 4, -3\rangle$ and $\mathbf{v} = \langle -1, 5\rangle$, compute: (a) $2\mathbf{u} - 3\mathbf{v}$, (b) the magnitude $||\mathbf{u}||$ and unit vector in the direction of $\mathbf{u}$, and (c) the dot product $\mathbf{u} \cdot \mathbf{v}$.

(a) Linear combination: $2\langle 4, -3\rangle - 3\langle -1, 5\rangle = \langle 8+3, -6-15\rangle = \mathbf{\langle 11, -21\rangle}$. (b) Magnitude and unit vector: $||\mathbf{u}|| = \sqrt{4^2 + (-3)^2} = \mathbf{5}$, $\hat{\mathbf{u}} = \left\langle \mathbf{\frac{4}{5}, -\frac{3}{5}} \right\rangle$. (c) Dot product: $\mathbf{u} \cdot \mathbf{v} = (4)(-1) + (-3)(5) = -4 - 15 = \mathbf{-19}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Vector arithmetic

Relative to a fixed origin O, the position vectors of the points A and B are given by $\vec{OA} = \mathbf{i} - 2\mathbf{j} + 4\mathbf{k}$ and $\vec{OB} = 4\mathbf{i} + 2\mathbf{j} - 8\mathbf{k}$. Find the unit vector in the direction of $\vec{AB}$.

a. $\frac{1}{13}(3\mathbf{i} + 4\mathbf{j} - 12\mathbf{k})$ ✓

b. $\frac{1}{13}(-3\mathbf{i} - 4\mathbf{j} + 12\mathbf{k})$

c. $\frac{1}{169}(3\mathbf{i} + 4\mathbf{j} - 12\mathbf{k})$

d. $\frac{1}{19}(3\mathbf{i} + 4\mathbf{j} - 12\mathbf{k})$

First, find the displacement vector $\vec{AB}$: $\vec{AB} = \vec{OB} - \vec{OA} = (4-1)\mathbf{i} + (2-(-2))\mathbf{j} + (-8-4)\mathbf{k} = 3\mathbf{i} + 4\mathbf{j} - 12\mathbf{k}$. Next, calculate the magnitude of $\vec{AB}$: $|\vec{AB}| = \sqrt{3^2 + 4^2 + (-12)^2} = \sqrt{9 + 16 + 144} = \sqrt{169} = 13$. The unit vector in the direction of $\vec{AB}$ is given by $\frac{\vec{AB}}{|\vec{AB}|}$: $\frac{1}{13}(3\mathbf{i} + 4\mathbf{j} - 12\mathbf{k})$. Common misconceptions: - Reversing the subtraction (cal.

22 [Foundational] • Target: Applications

Relative to a fixed origin O, the points A and B have position vectors $\vec{OA} = \mathbf{i} + 5\mathbf{j} - 2\mathbf{k}$ and $\vec{OB} = 3\mathbf{i} + 2\mathbf{j} + 4\mathbf{k}$. Find the length of the vector $\vec{AB}$.

a. 7 ✓

b. $\sqrt{31}$

c. $\sqrt{69}$

d. 49

To find the vector $\vec{AB}$, subtract the position vector of A from that of B: $\vec{AB} = \vec{OB} - \vec{OA} = (3-1)\mathbf{i} + (2-5)\mathbf{j} + (4-(-2))\mathbf{k} = 2\mathbf{i} - 3\mathbf{j} + 6\mathbf{k}$. The magnitude (length) of $\vec{AB}$ is: $|\vec{AB}| = \sqrt{2^2 + (-3)^2 + 6^2} = \sqrt{4 + 9 + 36} = \sqrt{49} = 7$. Common misconceptions: - Incorrectly taking $(-3)^2 = -9$ leads to $\sqrt{4 - 9 + 36} = \sqrt{31}$. - Adding th.

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23 [Foundational] • Target: Position vectors and geometry

The first, fourth, and sixth terms of an arithmetic progression (u_n) are equal to the first, second, and third terms, respectively, of an infinite geometric progression (v_n) with common ratio $re1$. Given that the sum to infinity of the geometric progression is 27, determine the maximum value of $S_n = \sum_{k=1}^{n} u_k$, the sum of the first n terms of the arithmetic progression.

a. 44

b. 45 ✓

c. 50

d. 54

Let the arithmetic progression (u_n) have first term a and common difference d. Then: $u_1 = a, \quad u_4 = a + 3d, \quad u_6 = a + 5d$. Let the geometric progression (v_n) have first term v_1 and common ratio r. Since $u_1 = v_1$, $u_4 = v_2$, and $u_6 = v_3$, we have: $v_1 = a, \quad v_2 = ar, \quad v_3 = ar^2$. Equating terms: 1) $a + 3d = ar \Rightarrow 3d = a(r-1) \Rightarrow d = \frac{a(r-1)}{3}$. 2) $a + 5d = ar^2$. Substitute d into the second equation.

24 [Foundational] • Target: More applications

In ΔABC, the side lengths are $AB = 9$, $AC = 8$, and $BC = 7$. A line segment $\overline{DE}$, with D lying on $\overline{AB}$ and E lying on $\overline{AC}$, divides ΔABC into two regions that have equal areas and equal perimeters. What is the length of $\overline{DE}$?

a. $2\sqrt{6}$ ✓

b. $\frac{7\sqrt{2}}{2}$

c. $4\sqrt{3}$

d. $6\sqrt{2}$

The line segment $\overline{DE}$ divides ΔADE into two regions: ΔADE and quadrilateral $BDEC$. 1. **Equal Perimeters:** The perimeter of ΔADE is $AD + AE + DE$. The perimeter of quadrilateral $BDEC$ is $DB + BC + CE + DE = (AB - AD) + BC + (AC - AE) + DE$. Equating the two perimeters and subtracting DE from both sides gives: $AD + AE = (AB + BC + AC) - (AD + AE)$ $2(AD + AE) = 9 + 7 + 8 = 24 \Rightarrow AD + AE = 12$ 2. **Equal Areas:** Using the sine formula for area.

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25 [Foundational] • Target: Finding direction angles

In trapezoid $ABCD$, base $\overline{AB}$ is parallel to base $\overline{CD}$, and $\angle DAB = \angle ADC = 90^\circ$. The length of $\overline{AB}$ is x, the length of $\overline{CD}$ is $2x + 1$, and the length of the non-parallel side $\overline{BC}$ is $2x$. If the height $AD = 8$, what is the area of trapezoid $ABCD$?

a. 34

b. **64** ✓

c. 80

d. 128

To find the area of trapezoid $ABCD$, we first determine the value of the parameter x. 1. **Deconstruct the trapezoid:** Draw an altitude from vertex B perpendicular to $\overline{CD}$, intersecting $\overline{CD}$ at point E. Because $\angle DAB = \angle ADC = 90^\circ$ and $\overline{BE} \perp \overline{CD}$, quadrilateral $ABED$ is a rectangle. Therefore: $BE = AD = 8 \quad \text{and} \quad DE = AB = x$ 2. **Express the sides of right triangle ΔBEC:** Point E lies on $\overline{CD}$, so the re.

26 [Foundational] • Target: Vector equations and unknowns

Determine the complete set of real values of the constant k for which the equation $e^x - kx^2 = 0$ has exactly three distinct real roots.

a. $k > \frac{e^2}{4}$ ✓

b. $k \geq \frac{e^2}{4}$

c. $0 < k < \frac{e^2}{4}$

d. $k > \frac{e}{2}$

First, note that $x = 0$ is not a solution to $e^x - kx^2 = 0$ since $e^0 = 1eq0$. Therefore, we can rewrite the equation as: $k = \frac{e^x}{x^2}$ Let $f(x) = \frac{e^x}{x^2}$ defined for all $xeq0$. The number of distinct real roots corresponds to the number of points of intersection between the curve $y = f(x)$ and the horizontal line $y = k$. Differentiating $f(x)$ using the quotient rule: $f'(x) = \frac{e^x \cdot x^2 - e^x \cdot 2x}{x^4} = \frac{e^x x(x-2)}{x^4} = \frac{e^x(x-2)}{x^3}$ Setting.

27 [Foundational] • Target: Parallel and equal vectors

The line l_1 has vector equation $\mathbf{r} = (1; 5; 0) + \lambda(1; -2; 2)$ and the line l_2 has vector equation $\mathbf{r} = (1; p; 2) + \mu(2; 3; q)$ where p and q are constants, and $\lambda, \mu \in \mathbb{R}$. Given that the lines l_1 and l_2 are perpendicular and intersect at a common point, find the value of p.

a. -2 ✓

b. 2

c. -4

d. 4

We solve the problem in three steps: **Step 1: Determine q using perpendicularity** The direction vectors of l_1 and l_2 are $\mathbf{d}_1 = (1; -2; 2)$ and $\mathbf{d}_2 = (2; 3; q)$. Since $l_1 \perp l_2$, their scalar (dot) product must equal zero: $\mathbf{d}_1 \cdot \mathbf{d}_2 = (1)(2) + (-2)(3) + (2)(q) = 0\ 2 - 6 + 2q = 0 \Rightarrow 2q = 4 \Rightarrow q = 2.$ **Step 2: Find the parameters λ and μ at the intersection** Equating the coordinates of l_1 and l_2.

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28 [Foundational] • Target: Vectors in three real-world scenarios

In a championship playoff, two teams, Team A and Team B, play a best-of-five series, where the first team to win three games wins the series. The games are independent, cannot end in a tie, and in each individual game, the probability that Team A wins is p, where $0 < p < 1$ and $p \neq q \frac{1}{2}$. Suppose the probability that the series is decided in exactly 3 games is equal to the probability that the series requires a decisive 5th game. What is the value of the product $p(1-p)$?

a. $\frac{\sqrt{33}-3}{12}$ ✓

b. $\frac{\sqrt{33}+3}{12}$

c. $\frac{\sqrt{33}-3}{6}$

d. $\frac{6-\sqrt{33}}{12}$

Let $q = 1 - p$ be the probability that Team B wins any individual game. 1. **Probability that the series ends in exactly 3 games:** This occurs if Team A sweeps $3-0$ or Team B sweeps $3-0$: $P(3 \text{ games}) = p^3 + q^3$ Using the algebraic sum of cubes identity: $p^3 + q^3 = (p+q)(p^2 - pq + q^2) = (p+q)((p+q)^2 - 3pq)$ Since $p + q = 1$, setting $x = pq = p(1-p)$ gives: $P(3 \text{ games}) = 1 - 3pq = 1 - 3x$ 2. **Probability that the series requires a 5th game:** A 5th game is played if and only if.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Vector arithmetic

The first three terms of a geometric progression are given by $u_1 = 3k - 2, \quad u_2 = k + 2, \quad u_3 = k - 2,$ where k is a real constant. Given that the infinite geometric series $\sum_{n=1}^{\infty} u_n$ converges, find the sum to infinity, S_∞.

a. 32 ✓

b. -1

c. $\frac{32}{3}$

d. 16

Step 1: Set up the common ratio property of a geometric sequence. Since the terms form a geometric progression, the ratio between consecutive terms must be equal: $\frac{u_2}{u_1} = \frac{u_3}{u_2} \Rightarrow u_2^2 = u_1 u_3$ Step 2: Substitute the expressions in terms of k: $(k+2)^2 = (3k-2)(k-2)$ $k^2 + 4k + 4 = 3k^2 - 8k + 4$ Step 3: Solve the quadratic equation for k: $2k^2 - 12k = 0 \Rightarrow 2k(k-6) = 0$ Thus, $k = 0$ or $k = 6$. Step 4: Apply the convergence criterion $|r| < 1$: - If $k = 0$.

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30 [Intermediate] • Target: Applications

Let $P(x)$ be a monic quartic polynomial with real coefficients whose graph is tangent to the x-axis at $x = -3$ and $x = 7$. A horizontal line $y = k$ intersects the graph of $y = P(x)$ at four distinct points whose x-coordinates, when listed in increasing order, form an arithmetic progression. What is the value of k?

a. 400 ✓

b. 225

c. 256

d. 625

1. **Determine $P(x)$:** Because $P(x)$ is a monic quartic polynomial that is tangent to the x-axis at $x = -3$ and $x = 7$, it has roots of multiplicity at least 2 at both points. Since $\deg(P) = 4$, these multiplicities must be exactly 2: $P(x) = (x + 3)^2(x - 7)^2$ 2. **Exploit the axis of symmetry:** The line of symmetry lies halfway between -3 and 7: $h = \frac{-3+7}{2} = 2$ Let $u = x - 2$. Then $x + 3 = u + 5$ and $x - 7 = u - 5$. In terms of u, the polynomial becomes.

31 [Intermediate] • Target: Position vectors and geometry

A vector $\mathbf{v}$ is defined by its component form $\langle 2a + 1, 3 - b \rangle$. This vector is then transformed by a translation represented by the vector $\mathbf{t} = \langle 5, -2 \rangle$. The resulting transformed vector, $\mathbf{v}'$, is given as $\langle 12, 1 \rangle$. Determine the values of the parameters a and b.

a. **a = 3, b = -4** ✓

b. a = 4, b = 3

c. a = 3, b = 4

d. a = 4, b = -3

The original vector is $\mathbf{v} = \langle 2a + 1, 3 - b \rangle$. The translation vector is $\mathbf{t} = \langle 5, -2 \rangle$. The transformed vector is $\mathbf{v}' = \langle 12, 5 \rangle$. The transformation is given by $\mathbf{v}' = \mathbf{v} + \mathbf{t}$. Substituting the given vectors: $\langle 12, 5 \rangle = \langle 2a + 1, 3 - b \rangle + \langle 5, -2 \rangle$ $\langle 12, 5 \rangle = \langle (2a + 1) + 5, (3 - b) + (-2) \rangle$ $\langle 12, 5 \rangle = \langle 2a + 6, 1 - b \rangle$ Equating the corresponding.

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32 [Intermediate] • Target: More applications

A solid cone has a height of 24 cm. A smaller cone is cut from the top of the larger cone by a plane parallel to the base. The smaller cone has a height of 8 cm. If the radius of the base of the larger cone is 12 cm, what is the radius of the base of the smaller cone?

a. **4 cm** ✓

b. 6 cm

c. 8 cm

d. 5 cm

Let H be the height of the larger cone and R be the radius of its base. Let h be the height of the smaller cone and r be the radius of its base. We are given $H = 24$ cm, $h = 8$ cm, and $R = 12$ cm. Since the smaller cone is cut by a plane parallel to the base, the smaller cone is similar to the larger cone. This similarity implies that the ratio of their corresponding linear dimensions is constant. Specifically, the ratio of the heights is equal to the ratio of the radii: $\frac{h}{H} = \frac{r}{R}$ Subst.

33 [Intermediate] • Target: Finding direction angles

The polynomial $P(x) = x^4 - 8x^3 + ax^2 + bx + c$ has four real roots that form an arithmetic progression. If the sum of the squares of the four roots is 36, what is the value of the constant term c?

a. -15 ✓

b. -9

c. 9

d. 15

Let the four real roots in arithmetic progression be represented symmetrically as $m - 3d$, $m - d$, $m + d$, and $m + 3d$, where $2d$ is the common difference. By Vieta's formulas, the sum of the roots is the negative of the coefficient of x^3: $(m - 3d) + (m - d) + (m + d) + (m + 3d) = 4m = 8 \Rightarrow m = 2$. Next, the sum of the squares of the roots is given as 36: $(2 - 3d)^2 + (2 - d)^2 + (2 + d)^2 + (2 + 3d)^2 = 36$. Expanding each term.

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34 [Intermediate] • Target: Vector equations and unknowns

A cone has a height of 18 cm. A smaller cone is formed by slicing the top of the original cone with a plane parallel to the base. The smaller cone is similar to the original cone, and its height is 12 cm. If the radius of the base of the original cone is 9 cm, what is the radius of the base of the smaller cone?

a. 6 **cm** ✓

b. 5 cm

c. 4.5 cm

d. 3 cm

Let the height of the original cone be H and its radius be R. Let the height of the smaller cone be h and its radius be r. We are given that $H = 18$ cm, $R = 9$ cm, and $h = 12$ cm. Since the smaller cone is similar to the original cone, the ratio of their corresponding linear dimensions is constant. Therefore, the ratio of their heights is equal to the ratio of their radii: $\frac{h}{H} = \frac{r}{R}$ Substituting the given values: $\frac{12}{18} = \frac{r}{9}$ To solve for r, we can simplify the fracti.

35 [Intermediate] • Target: Parallel and equal vectors

A right circular cone has a height of 12 cm. A smaller cone is formed by slicing the top of the larger cone with a plane parallel to the base. The smaller cone is similar to the larger cone, and its height is 6 cm. If the radius of the base of the larger cone is 9 cm, what is the radius of the base of the smaller cone?

a. 4.5 **cm** ✓

b. 5 cm

c. 6 cm

d. 7.5 cm

Let the larger cone be denoted by C_1 and the smaller cone by C_2. The height of C_1 is $H_1 = 12$ cm and its radius is $R_1 = 9$ cm. The height of C_2 is $H_2 = 6$ cm. Since the smaller cone is similar to the larger cone, the ratio of their corresponding linear dimensions is constant. Therefore, the ratio of their heights is equal to the ratio of their radii. $\frac{R_2}{R_1} = \frac{H_2}{H_1}$ Substitute the given values: $\frac{R_2}{9\text{ cm}} = \frac{6\text{ cm}}{12\text{ cm}}$.

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36 [Intermediate] • Target: Vectors in three real-world scenarios

In the diagram below, lines DE and BC are parallel. If AD = 6, DB = 4, and AE = 9, what is the length of EC?

a. **6** ✓

b. 7

c. 8

d. 10

Since DE is parallel to BC, triangle ADE is similar to triangle ABC. By the properties of similar triangles, the ratio of corresponding sides is equal. Therefore, we have the proportion: $\frac{AD}{AB} = \frac{AE}{AC}$. We are given AD = 6, DB = 4, and AE = 9. So, AB = AD + DB = 6 + 4 = 10. Substituting these values into the proportion: $\frac{6}{10} = \frac{9}{AC}$. To find AC, we can cross-multiply: $6 \times AC = 10 \times 9$ $6 \times AC = 90$ $AC = \frac{90}{6}$ $AC = 15$. We are asked to find the length of EC.

37 [Intermediate] • Target: Vector arithmetic

A vector **v** is defined by its initial point $P = (-5, 3)$ and its terminal point $Q = (4, -2)$. What is the component form of vector **v**?

a. $\langle -9, 5 \rangle$

b. $\langle 9, -5 \rangle$ ✓

c. $\langle -1, 1 \rangle$

d. $\langle 1, -1 \rangle$

The component form of a vector **v** directed from an initial point $P = (x_1, y_1)$ to a terminal point $Q = (x_2, y_2)$ is given by $\mathbf{v} = \langle x_2 - x_1, y_2 - y_1 \rangle$. In this case, the initial point is $P = (-5, 3)$ and the terminal point is $Q = (4, -2)$. Therefore, the component form of **v** is:
$\mathbf{v} = \langle 4 - (-5), -2 - 3 \rangle = \langle 4 + 5, -5 \rangle = \langle 9, -5 \rangle$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Applications • Cross-Lesson: 2.01 Right-Triangle Trigonometry

Consider the vector $\mathbf{v}$ with initial point $P = (2, 3)$ and terminal point $Q = (5, 7)$. Find the component form of $\mathbf{v}$.

a. $\langle 3, 4 \rangle$ ✓

b. $\langle 7, 10 \rangle$

c. $\langle -3, -4 \rangle$

d. $\langle 4, 3 \rangle$

The component form of a vector $\mathbf{v}$ with initial point (x_1, y_1) and terminal point (x_2, y_2) is given by $\mathbf{v} = \langle x_2 - x_1, y_2 - y_1 \rangle$. In this case, $(x_1, y_1) = (2, 3)$ and $(x_2, y_2) = (5, 7)$. Therefore, the component form of $\mathbf{v}$ is $\langle 5 - 2, 7 - 3 \rangle = \langle 3, 4 \rangle$.

39 [Challenging] • Target: Position vectors and geometry • Cross-Lesson: 2.02 Radian Measure and Arc Length

A transformation maps point $A = (1, 2)$ to $A' = (-4, 2)$. If this transformation is a translation, what is the component form of the translation vector $\mathbf{v}$?

a. $\langle -5, 0 \rangle$ ✓

b. $\langle 5, 0 \rangle$

c. $\langle -3, 0 \rangle$

d. $\langle 3, 0 \rangle$

Let the initial point be $A = (x_1, y_1)$ and the translated point be $A' = (x_2, y_2)$. The translation vector $\mathbf{v}$ is given by $\mathbf{v} = \langle x_2 - x_1, y_2 - y_1 \rangle$. In this case, $(x_1, y_1) = (1, 2)$ and $(x_2, y_2) = (-4, 2)$. Therefore, $\mathbf{v} = \langle -4 - 1, 2 - 2 \rangle = \langle -5, 0 \rangle$.

40 [Challenging] • Target: More applications • Cross-Lesson: 2.03 Graphs of Sine and Cosine Functions

Consider the transformation grid below. If the original point A is mapped to A', what is the vector representing this translation?

a. $\langle -5, 0 \rangle$

b. $\langle -6, 0 \rangle$ ✓

c. $\langle 5, 0 \rangle$

d. $\langle 6, 0 \rangle$

The vector representing a translation from point $P = (x_1, y_1)$ to point $P' = (x_2, y_2)$ is given by $\langle x_2 - x_1, y_2 - y_1 \rangle$. In this case, point A is at $(1, 2)$ and point A' is at $(-5, 2)$. Therefore, the translation vector is $\langle -5 - 1, 2 - 2 \rangle = \langle -6, 0 \rangle$.

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41 [Challenging] • Target: Finding direction angles • Cross-Lesson: 2.04 Law of Sines and Law of Cosines

Consider the vector $\mathbf{v}$ directed from the point $P = (2, 5)$ to the point $Q = (7, 1)$. What is the component form of $\mathbf{v}$?

a. $\langle -5, -4 \rangle$

b. $\langle 5, 4 \rangle$

c. $\langle 5, -4 \rangle$ ✓

d. $\langle -5, 4 \rangle$

The component form of a vector directed from point $P = (x_1, y_1)$ to point $Q = (x_2, y_2)$ is given by $\mathbf{v} = \langle x_2 - x_1, y_2 - y_1 \rangle$. In this case, $P = (2, 5)$ and $Q = (7, 1)$. Therefore, the component form of $\mathbf{v}$ is $\langle 7 - 2, 1 - 5 \rangle = \langle 5, -4 \rangle$.

42 [Challenging] • Target: Vector equations and unknowns • Cross-Lesson: 3.01 The Unit Circle and Exact Values

A particle in equilibrium is acted on by three concurrent forces. Two of the forces have equal magnitude 20 N and act symmetrically either side of the third force's line of action. If the angle between each of the two equal forces and the third force is 120°, find the magnitude of the third force.

a. 18

b. **20** ✓

c. 25

d. 22

By symmetry (each pair at 120°), the third force balances the resultant of the two equal forces: $F_3 = F_1 = 20$ N (standard 120°-120°-120° symmetric equilibrium).

43 [Challenging] • Target: Parallel and equal vectors • Cross-Lesson: 3.02 Fundamental Trigonometric Identities

Find the real number(s) equal to $(-32)^{3/5}$.

a. 8

b. Undefined

c. -8 ✓

d. -32^3

Applying Radicals & Rational Exponents principles, $(-32)^{1/5} = -2$. $(-2)^3 = -8$. Therefore, the verified correct option is -8.

44 [Challenging] • Target: Vectors in three real-world scenarios • Cross-Lesson: 3.03 Sum and Difference Formulas

What are the equations of the asymptotes of the hyperbola $\frac{(x-1)^2}{9} - \frac{(y+2)^2}{4} = 1$?

a. $y + 2 = \pm\frac{2}{3}(x - 1)$ ✓

b. $y + 2 = \pm\frac{3}{2}(x - 1)$

c. $y = \pm\frac{2}{3}x$

d. $y - 2 = \pm\frac{2}{3}(x + 1)$

Center $(h, k) = (1, -2)$, slopes $\pm\frac{b}{a} = \pm\frac{2}{3}$. Asymptotes: $y + 2 = \pm\frac{2}{3}(x - 1)$.

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45 [Challenging] • Target: Vector arithmetic • Cross-Lesson: 3.04 Double-Angle Formulas and Trigonometric Equations

The three angles of a triangle measure $(3x - 5)^\circ$, $(3x - 10)^\circ$, and $(x + 20)^\circ$. Find the measure of the largest angle.

a. 70 ✓

b. 64

c. 62

d. 65

$(3x - 5) + (3x - 10) + (x + 20) = 180 \Rightarrow x = 25$. Angles: 70°, 65°, 45°; largest $= 70^\circ$.

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